

Final Project Proposal

Parking Lot Entry Booth

CSS142P Modelling and Simulation | School of Information Technology, Mapúa University

Field	Entry
Working title	Single-booth vehicle entry queue at the Mapúa University Makati campus parking lot
Group number	8
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The system in one line	Vehicles arrive at the campus parking lot's single entry booth, queue along the entry approach, are processed one at a time by an attendant who issues a ticket or checks an ID, and proceed into the lot once cleared.
Part 1: problem and decision	The lot is served by one entry booth staffed from 07:00 to 19:00. On high-traffic mornings the entry queue grows past the gate and onto Pablo Ocampo Sr. Extension, where waiting vehicles block a lane of through traffic and draw complaints from drivers and neighbouring establishments. Management must decide whether the cost of an additional attendant during the morning peak is justified by the reduction in queueing. The decision is: staff a second entry booth from 07:00 to 09:00 on high-traffic days, or keep a single booth for the whole day.
Part 2: entities	Vehicles arriving to enter the lot. One vehicle is one entity, for example, a faculty car that arrives at 07:42, waits behind six other vehicles, is issued a ticket, and enters.
Part 2: resources	Entry booths, each processing one vehicle at a time. Baseline configuration: one booth, open 07:00–19:00. Alternative configuration: a second booth open 07:00–09:00, then one booth for the remainder of the day.
Part 2: arrivals	Vehicles arrive one at a time across the 12-hour operating day. Arrivals are modelled as a Poisson process whose rate is piecewise constant over 15-minute intervals, so the 07:00–09:00 morning peak is represented by a higher arrival rate in those intervals rather than by a separate mechanism.

Part 2: run start and stop	A run is one operating day, 07:00 to 19:00. Arrivals stop at 19:00; vehicles already in the queue at that point are processed before the run ends. Each run starts with an empty queue and an idle booth.
Part 2: out of scope	1. Parking space availability inside the lot where every cleared vehicle is assumed to find a slot. 2. The exit booth and departure traffic. 3. Payment disputes, lost tickets, and manual supervisor overrides. 4. Balking (drivers who see the line and go elsewhere) and reneging (drivers who join the queue then leave) every arriving vehicle joins the queue and stays until processed.
Part 3: model type	Discrete-event simulation.
Part 3: why this type fits	The state variables are the number of vehicles waiting in the queue and the busy-or-idle status of each booth. Both change only at discrete instants and hold constant in between, so there is no continuously varying level to integrate. The event set is: vehicle arrival, processing start, processing completion, and the closing of the second booth at 09:00 in the two-booth configuration.
Part 4: measures, units, computation	Average waiting time (minutes): mean across all vehicles of processing-start time minus arrival time. Maximum queue length (vehicles): running maximum of the number of vehicles waiting, sampled at every event. Booth utilization (fraction of staffed time): total busy booth-time divided by total booth-time staffed, so that the one-booth and two-booth configurations remain comparable even though the second booth is open for only two hours. Spillover duration (minutes): total simulated time during which the queue holds at least K vehicles, where K is the number of vehicles the entry approach can hold before the tail of the queue reaches Pablo Ocampo Sr. Extension.
Part 5: inputs and where each value comes from	Arrival counts per 15-minute interval: taken from the main gate entry log for one full high-traffic operating day, using the recorded entry timestamps to reconstruct arrivals per interval. Per-vehicle processing time: stopwatch timing of 30 vehicles during the 07:00–09:00 peak window and 30 vehicles off-peak. A distribution (exponential, lognormal, or triangular) will be fitted to the sample and the chosen fit reported; no distribution is assumed in advance. Approach capacity K (vehicles): measured on site before the model is built, as the usable approach length divided by average vehicle length plus inter-vehicle gap. Booth count and staffing hours: campus policy, known exactly (current = 1 booth, 07:00–19:00; alternative = 2 booths during the peak window).

	Assumption: the second booth is assumed to draw processing times from the same fitted distribution as the first.
Part 6: tools	Python 3.11, SimPy, and NumPy for the model; pandas for summarising the observed input data. Thirty independent replications per configuration, using common random numbers across the one-booth and two-booth runs so that the difference between them reflects the staffing change rather than seed noise. All random seeds are recorded.